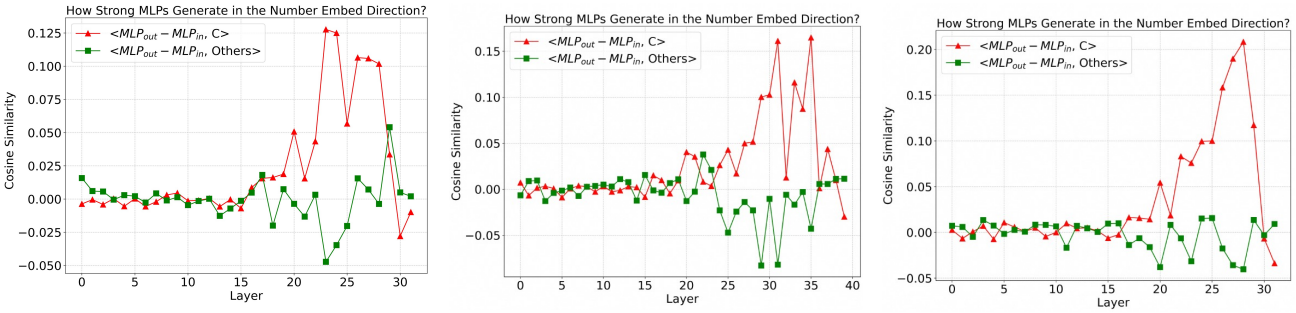


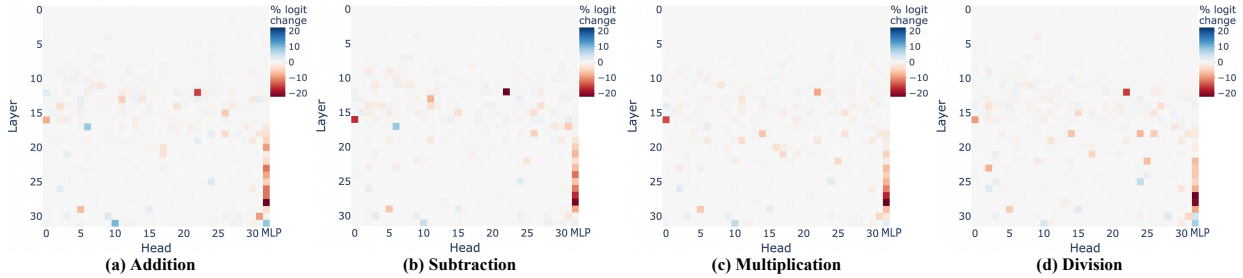
(a) LLaMA2-7B (b) LLaMA2-13B (c) Mistral-7B

Figure 14: We investigate the projection of each MLP layer (MLP_{out}-MLP_{in}) along the direction of number token {C} (*i.e.*, right answer) and other tokens (*i.e.*, wrong answer).

$inflection\text{-}end$), are as follows: (13-18-28) for LLaMA2-7B, (13-18-35) for LLaMA2-13B, and (13-20-28) for Mistral-7B. In Figure 14, the similarities of MLP_{out}-MLP_{in} and right answer {C} show a pattern of initial stabilization followed by an increase. The critical points for LLaMA2-7B/LLaMA2-13B/Mistral-7B are again (13-18-28), (13-18-35), and (13-20-28). The inflection points in both Figure 13 and Figure 14 are nearly identical, indicating consistent trend shifts across the models. It helps to verify that LLMs initially leverage attention heads then relaying information to downstream MLPs, to progressively carry out the calculation to final results. Furthermore, the above findings appear to be general and robust across different LLMs, not limited to a specific model.

D. Key Component Location across Calculation Tasks.

We investigate the location of key components for each calculation task individually, as shown in Figure 15. The discovered key heads could be shared across four tasks, which are sparsely distributed in the middle layers. Specifically, when examining subtraction and addition tasks, we could summarize two insightful symmetries between them. The identified key heads of two tasks are almost the same, albeit with different magnitude of the effect. This phenomenon could reveal the symmetry of key head "*location*" in addition and subtraction. Moreover, the tasks of multiplication and division exhibit a greater number of key heads compared to the tasks of addition and subtraction. We assume it could be attributed to their more intricate operations within multiplication and division.



(a) Addition (b) Subtraction (c) Multiplication (d) Division

Figure 15: We conduct path patching experiments on LLaMA2-7B across four mathematical tasks, by searching for each head and MLP directly affecting the logit of the right answer. The last column denotes the path patching results of MLPs. For each head/MLP, a darker color indicates a larger logit difference from the original model before patching.

Generalize to other calculation operations. We conduct the experiments of key head identification and validation following Section 5.1. We generate the samples including the exponentiation operation as the reference data X_{r}. Then we generate the counterfactual data X_{c} following the principles introduced in Section 4.1 to exclude the exponentiation logic.

The results reveal the potential of generalizing to more complex mathematical operations: (i) Five key heads are identified based on the newly generated X_{r} and X_{c}. We find that the heads (11, 8) and (14, 2) mainly attend to the operators "^", "power", while the heads (12, 22), (13, 11), (15, 15) mainly attend to the input operands {A} and {B}. (ii) Knocking out the key heads, identified by both templates, leads to significantly impacts (over 60%) on model performance.

Table 3: Key head identification on the exponentiation task.

Templates	Key Heads [*e.g.*, (Layer, Head)]	Knockout Accuracy
X_{r}: "{A} $\wedge$ {B} = _ " X_{c}: "{A} $<$ {B} = _ "	[(11, 8), (12, 22), (13, 11), (14, 2), (15, 15)]	-66%
X_{r}: "{A} to the **power** of {B} equals _ " X_{c}: "{A} to the **none** of {B} equals _ "	[(11, 8), (12, 22), (13, 11), (14, 2), (15, 15)]	-62%

E. Generalize to More Complex Scenarios.

We conduct experiments on the more complex scenario using the dataset GSM8K (Cobbe et al., 2021). At first, we create new reference data X_{r} and counterfactual data X_{c}. Following the idea of methodology proposed in Section 4.1, we convert the question in GSM8K to obfuscate the semantic elements that necessitate calculation, while ensuring that the alterations to the text are minimal. An example is shown below:

- GSM8K X_{r}: "On a 16 GB (gigabyte) capacity USB drive, 50% is already busy. **Calculate the number** of gigabytes still available."
- GSM8K X_{c}: "On a 16 GB (gigabyte) capacity USB drive, 50% is already busy. **Describe the location** of gigabytes still available."

Then, we conduct the experiments of key head identification and validation following the experimental setting in Section 5.1. As a result, 60% of the key heads are overlapped with the key heads identified based on our original data. Moreover, knocking out the newly-identified key heads leads to a 65% accuracy drop on GSM8K, confirming their importance even in complex scenarios.

Table 4: Comparison of the key heads identified on our generated data in Figure 8 and the dataset GSM8K (Cobbe et al., 2021).

Dataset	Top-10 Key Heads [*e.g.*, (Layer, Head)]	Knockout Accuracy
Ours	[(12, 22), (13, 11), (16, 0), (15, 26), (18, 26), (18, 24), (30,31), (14, 27), (22, 25), (11, 8)]	-69%
GSM8K	[(19, 6), (11, 8), (12, 22), (14, 31), (13, 11), (22, 25), (16, 0), (21, 17), (15, 26), (29, 5)]	-65%

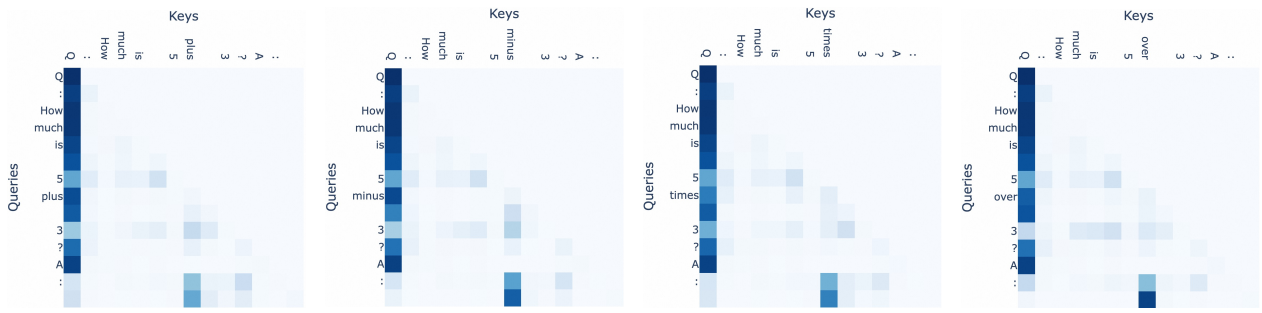
Furthermore, only knocking out the 6 overlapping heads brings in -56% and -52% on our generated data and GSM8K, respectively. It shows these heads are both important in two scenarios. If knocking out the 4 non-overlapping heads identified by GSM8K only, it has a negligible effect on our generated data (-2%) but apparently affects on GSM8K (-26%). It reveals the significance of these 4 heads specific to more complex reasoning mathematical problems. We further investigate the attention patterns of the 4 non-overlapping heads, and find that these heads mainly attend to text tokens. For example, the head $(29, 5)$ attends to ".", and the head $(19, 6)$ attends to "GB". In contrast, the 6 overlapping heads mainly attend to the number operands and operators. For example, the head $(13, 11)$ attends to input operands "50", and the head $(11, 8)$ attends to the operator "%".

Recent research (Opedal et al., 2024) has shown that solving the math word problems requires a synergy of multiple skills including 'text comprehension' and 'arithmetic calculation'. This is aligned with the phenomena of "the 4 non-overlapping heads attend to text tokens (*i.e.*, 'text comprehension'), while the 6 overlapping heads attend to number operands and operators (*i.e.*, 'arithmetic calculation')". In this work, we focus on the skill of arithmetic calculation as it's a fundamental ability universally shared across various levels of complexity for mathematical problems. It's imperative for continued research to develop a more holistic understanding of the intricate reasoning capacities.

To further investigate whether the model's deficiencies stem from a lack of mathematical abilities or a broader impairment in language processing, we evaluate LLaMA2-7B with key heads kept normal and knocked out on MMLU-Humanities benchmark (Hendrycks et al., 2020). The comparative performance was 42.9% for models with the key heads intact versus 42.6% for the knockout models. This negligible difference (-0.3%) suggests that the knockout of these heads does not significantly impact general language abilities.

F. More Attention Pattern Cases.

We show the attention patterns of the operator-attended heads (*e.g.*, 14.2) in Figure 16 that could attend to the tokens of "plus", "minus", "times", and "over", across different sentences.



(a) Head 14.2 (b) Head 14.2 (c) Head 14.2 (d) Head 14.2

Figure 16: The attention patterns of the key head 14.2, which mainly attend to the operator-related tokens, *e.g.*, "plus", "minus", "times", "over".

G. Ablation Study of Precise SFT on MLPs.

We further investigate the influence of different number of tuned MLPs in Table 5. It reveals that the tuning more MLPs could lead to a performance decrease on MMLU and more training time, while imrove the performance on math dataset GSM8K.

Table 5: Ablative experiments on the number of tunable MLPs.

Precise SFT Setting	**Evaluation Metric**			
	Train Speed	Tunable Params.	GSM8K	MMLU
top-32 heads	50sam./sec.	0.067B	27.4	46.4
top-32 heads + top-1 MLP	44sam./sec.	0.202B	27.5	46.0
top-32 heads + top-2 MLPs	38sam./sec.	0.338B	27.7	45.7
top-32 heads + top-3 MLPs	31sam./sec.	0.473B	28.0	45.2
top-32 heads + top-6 MLPs	26sam./sec.	0.879B	28.2	44.9
top-32 heads + all MLPs	19sam./sec.	4.396B	29.2	43.9

H. Calculation in Computer vs LLMs.